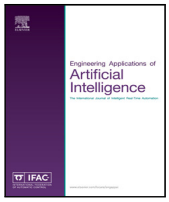
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Hierarchical concept Bottleneck models for vision and their application to explainable fine classification and tracking



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ABSTRACT

Improving the explainability of Computer Vision models based on Deep Learning has recently become a compelling problem, ensuring reliable predictions to the end-user and enabling more fine-grained classifications. Recently, Concept Bottleneck models have been proposed for images classification, partitioning the problem in two stages and thereby defining a hierarchy of concepts. So far, however, little work has been done to investigate the applicability of this approach to other datasets with higher intra-class variability and ambiguity, and to discuss its flexibility to tasks different from whole-images classification. In this work we develop and discuss a Concept Bottleneck model for images segmentation, objects fine classification and tracking, and compare it to more classical methods based on Mask R-CNN and images similarity algorithms. All our models are trained and tested on a dataset comprised of pictures of fridges filled with various objects, however the method can be applied to any fine classification task. The proposed model makes full use of the hierarchy in concepts, exploiting the relationships between different categories at the same hierarchical level and relying on a novel method for handling multi-labels classifications. We show that the performance on fine classification is on par with a regular Mask R-CNN, but with a significant increase in explainability and in handling classes confusion. New explainable metrics are proposed to quantitatively evaluate the increase in explainability. We also demonstrate the effectiveness of the derived Concept Bottleneck features on related tasks, i.e., the tracking of objects between consecutive pictures in a sequence. The code is released as open source and available at https://opensource.silicon-austria.com/pittinof/hierarchical-concept-bottleneck.

1. Introduction

Images segmentation and objects classification are typical problems in Machine Vision, which have been very successfully addressed by Deep Learning methods, such as Mask R-CNN (He et al., 2017). The most popular models, however, are usually treated as a "black-box", limiting the explainability of their predictions. This can be especially troublesome when the classification needs to be very fine, i.e., when having to differentiate between very similar objects, since in this case usually minor differences between objects are relevant. We believe that understanding why classification errors occur can be helpful in improving the performance in objects fine classification.

Recently, Concept Bottleneck models, Koh et al. (2020), have been proposed, with the purpose of improving the explainability of the classification output. Instead of directly classifying an image into the target classes, these models work in two stages: in the first one the so-called concepts are predicted, and in the second stage these are used for the final classification. Concepts can be thought of as object categories or object attributes, that describe the image features similarly to how humans would approach classification tasks. These concepts constitute then the lower level in a hierarchy of classes, and are directly used to predict the classes at a higher level. Using concepts, even in cases where the final label is wrongly classified, one can easily find explanations. Moreover, an intervention to correct the wrongly predicted concepts can be performed, which could then improve the overall performance of the model on the final classification. On the other hand, in cases where the model is not certain about the higher-level class of an image, for example because the object had not been seen before, the information extracted from the low-level concepts can still provide valuable information. All of these advantages can come, however, at the cost of a reduction in accuracy, due to the potential loss of information with the concepts abstraction, and of an increase in annotations effort. Therefore, a balance between explainability, annotations effort and accuracy might be necessary.

In this paper we investigate the usage of such hierarchical Concept Bottleneck models for images segmentation and fine objects classification, also introducing a novel way of handling multi-labels classification which we believe is more advantageous for such models. Moreover, we demonstrate another advantage of Concept Bottleneck models by adapting them for object tracking, and we compare our models to more traditional ones based on Mask R-CNN and images similarity features. Our experiments were performed on a proprietary dataset, derived from the use-case of smart fridges. It consists of pictures of fridges filled with various objects, with varying point of view and conditions. Each recognized object should be finely classified, a very complex task given the wide nature and diverse visual appearances of the available food items. In addition, some of the images are taken in

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a sequence, mimicking the behaviour of a user that repeatedly opens the fridge and displaces, removes or adds objects. Object tracking on such image sequences is different from conventional object tracking in videos, therefore state-of-the-art methods cannot be applied in this scenario. Our approach for objects tracking relies, instead, on image-to-image comparisons between the objects cutouts in two consecutive images, making use of the developed concepts. Such an approach is, therefore, more similar to the ones used for images similarity and image retrieval than the ones commonly associated with objects tracking.

The paper is organized as follows: Section 2 discusses in detail the related work in this field. Section 3, instead, presents the details of the dataset and of the developed models for fine classification and object tracking, discussing also our new approach for multi-labels classification. Section 4 presents the results obtained for the two applications on our dataset, and finally Section 5 draws the conclusions.

2. Related work

Multi-labels classification is an extension of simple classification in which multiple labels need to be assigned to each sample. Following the success of convolution neural network (CNN) models for image classification, several methods have been proposed (Wei et al., 2016; Wang et al., 2016; Zhu et al., 2017; Lee et al., 2018; Chen et al., 2019) to extend the approach from single to multi-label. As first suggested in Madjarov et al. (2012), problem transformation methods can be used, thereby separating the problem into multiple independent binary classification problems or designing a multi-class classification problem using all combinations of labels. However, transforming the problem to multiple binary classifiers fails to consider any relation between the labels and also shows redundancy by repeatedly extracting common features. On the other hand, the solution presented in Jia et al. (2020) reduces the redundancy in feature extraction using transfer learning, but it fails to consider the dependencies between the labels. Other recent approaches for multi-labels image classification incorporate the relationship between the labels in graph-based structures (Lee et al., 2018; Chen et al., 2019), while another method uses an encoder transformer trained to predict the target labels by using masked label embeddings and image features (Lanchantin et al., 2021). In Dai et al. (2020) a feature correlation learning network is proposed to exploit feature correlations and predict a binary vector of indicators for each label. A similar network structure is used as part of the model proposed in Yan et al. (2019), that learns the label dependencies indirectly by learning the correlations between the convolutional features.

Since most of the previous work focused on image-level predictions, without segmentation of individual objects, their goal was to predict a set of labels that correspond to different objects, all present in a single image. Our work instead focuses on single objects that are extracted from images through an image segmentation algorithm. Therefore, our goal is to predict labels that correspond to multiple features, concepts or attributes of every single object shown in the image. This scenario, recently denominated a concept-oriented problem (Jia et al., 2020), is still relatively unexplored. Although both spatial and semantic relationships between labels can be observed in both scenarios, an important difference is that the semantic relationship between labels is predominant in the case of the concept-oriented problem. One solution to the concept-oriented problem in multi-labels image classification is applied to a human attribute recognition task where the labels represent attributes related to gender, hair style, age, etc. Jia et al. (2020), while another is used to label product images with labels related to category, colour, form and other design-related labels (Dai et al., 2020). However, both solutions use binary attributes for each label, which allows for impossible combinations and fails to consider exclusions between different classes of the same category and relations between different categories.

Moreover, most research in the area of image classification has so far focused on the generic classification of objects to coarse-grained meta-categories. However, in recent years, several deep learning methods for fine-grained image classification have been explored, such as Wei et al. (2021), addressing the classification of objects from a meta-category to multiple sub-categories. The greatest challenge in fine image classification comes from small inter-class variations between different sub-categories and large intra-class variations between objects in the same sub-category. According to Wei et al. (2019), image classification systems still cannot achieve excellent performance, requiring the inclusion of external data such as text, web data, etc. to improve accuracy. Some solutions suggest, instead, using a loss function tailored for the fine classification (Dubey et al., 2018; Chang et al., 2020). The model proposed in Peng et al. (2018) improves multi-view and multi-scale feature learning by introducing a part-level attention model, which aims to select the discriminative parts of the objects, together with an object-level attention model, which aims to select the objects in the images.

Despite these advancements, the largest shortcoming of such Deep Learning models used for fine image classification is that they cannot explain their output in a way understandable and reconstructable by humans (Goebel et al., 2018). The approach followed by Chen et al. (2019) aims at explaining the classification output of a partly interpretable neural network by showing some evidence from the parts of the image that lead to the prediction. Furthermore, Wang et al. (2021) proposes a network with transparent basis concepts that can explain what concepts the model can learn and how the evidence is combined to make the classification. Other approaches make use of some domain knowledge to construct the classification features (Li and Yang, 2021), or develop hierarchical classifiers based on handmade features (Kliangsuwan and Heednacram, 2018), both approaches having the disadvantage of requiring intensive engineering and deep domain knowledge. On the other hand, in our work, we decided to work with a list of predefined concepts for classification, similarly to the approach in Koh et al. (2020). In this way, the model's prediction can be easily explained in terms of defined concepts, and an additional level of fine-tuning is available to improve the overall accuracy.

Finally, as discussed in the previous section, our approach for object tracking, due to the specificity of the studied application, leans more towards the field of image similarity. While general similarity, as suggested in Appalaraju and Chaoji (2017), involves identifying whether two images look similar, a special problem is identifying whether two images contain the same object. The latter problem has been studied in popular facial recognition systems (Schroff et al., 2015), but also in applications for person re-identification (Yi et al., 2014; Filković et al., 2016) and vehicle re-identification (Zhu et al., 2018), where the task is to match people or vehicles recorded with cameras from different points of view. State-of-the-art solutions involve Deep Learning models that can extract image embeddings trained to reflect the distance between images in such a way that this distance is minimized between the matching images and maximized between the mismatching images. One popular solution is the Siamese network (Yi et al., 2014), which uses a pair of matching or non-matching images as input, and another is the Triplet network (Hoffer and Ailon, 2015) which uses three images as input, where two of them are a match and the last one is a mismatch. Recently, Chen et al. (2020) proposed an approach, along with a new contrastive loss, in which images are taken as input in pairs, similarly to the Siamese network. Two networks are learned in parallel: a base encoder, that encodes the image representation, and a projection head, where the image representation is transformed to a space where the contrastive loss is calculated. Next, in Yu et al. (2018), a solution for image matching is proposed that uses the CNN feature pyramid, which is a hierarchical framework that detects patterns at different levels. Similar models to the ones used for image similarity are also used for image retrieval tasks, Shimoda and Yanai (2017), Ciocca et al. (2018) and Chaudhuri et al. (2019). For example, the Triplet network showed good performance in food image retrieval using a dataset with 256 fine-grained food categories, where images from



Fig. 1. Dataset examples.

the same category are considered similar to each other (Shimoda and Yanai, 2017). Finally, the features extracted from a fine-tuned ResNet-50 showed robust performance for fine-grained food classification and retrieval, where a database with 475 food classes is used (Ciocca et al., 2018).

Our approach, on the other hand, aims at using the features defined by the Concept Bottleneck model as input to the classifier, in the place of images embeddings. Such an approach has the advantages of both being much more explainable than state-of-the-art models, and of enabling to directly reuse the output of the Concept Bottleneck model, improving the computational efficiency.

In summary, the novel contributions of our paper are:

- definition of a new method for multiple-labels classification, more suited for the use with a Concept Bottleneck approach and that avoids incompatible classes combinations;
- development of a new explainable model, based on the Concept Bottleneck approach, for fine classification of objects;
- proposal of new metrics for the evaluation of the explainability of the developed models;
- development of a new explainable object tracking method, that reuses the features computed by the Concept Bottleneck model.

3. Methods

3.1. Dataset

The dataset consists of 1006 images of fridges filled with various products, as shown in Fig. 1. Of these images, 439 belong to 9 sequences, collected in such a way that only small changes are introduced between one picture and the next one in the sequence (for instance only a couple of objects are changed, some are moved or slightly displaced, see Fig. 2), thereby mimicking the behaviour of a normal use of the fridge. The remaining images, instead, present various situations of fridges, from full to relatively empty, with different kinds of objects inside. The images can have different sizes, aspect ratios and resolutions, and are acquired with different types of cameras. The point of view of the camera is always located on the fridge door, but the exact position varies between the pictures, and it is never exactly constant even inside a sequence.

The images have been annotated with contours and categories for each object. It has been chosen not to annotate objects that are barely



Fig. 2. Example of three consecutive images in a sequence.

visible, for example due to poor lighting (see top right image in Fig. 1), while all partially visible objects have been annotated. On average, each image consists of 10–11 objects, but this can vary considerably. To allow for object tracking, each different product has been assigned a unique identifier, thereby allowing one to identify the location of a specific product in all images in a sequence. Note that the overhead on the annotations process due to the multiple categories is minimal since:

- The most complex task of a human annotator involves drawing the contours for each object, and this part is independent of the number of categories.
- Once the contours are drawn, assigning one or multiple tags to a region does not create a significant overhead.
- If, during the dataset acquisition, as in this paper, a database of unique objects with their identifiers is generated, the annotator only needs to assign this identifier to each region and all tags can be automatically extracted from the database.

The dataset has been partitioned in two different ways, depending on the goal:

1. For objects classification, 4 out of the 9 sequences have been chosen mainly for test, using 90% of the images in the sequence, while all other images are randomly split between a training and a validation set, with a ratio of 5:1.
2. For objects tracking, only the sequences data are used. Due to the limited amount of data in this case, only training and test sets have been defined, with the latter including all images from 4 sequences.

3.2. Multiple-labels classification

Given the very specific nature of the objects of interest in this work, all belonging to food or beverage classes, the need of defining various attributes for the objects has been identified. Moreover, defining these attributes as a hierarchy of concepts, as illustrated in Fig. 3, enables a better specification of the object's characteristics and allows for an easier fine classification. In this work, the attributes have been defined with a two-fold purpose:

- the lower level concepts are designed as more general concepts, thereby easier for a classifier to predict; in this way even if the object had never been seen before, some information can still be extracted.
- the prediction of higher level concepts can be directly derived from the low-level concepts using the Concept Bottleneck approach (Koh et al., 2020), thereby providing a better explainability.

The low-level tags have been defined as follows. The most general category, called 3D shape (Table 1), aims at defining the general

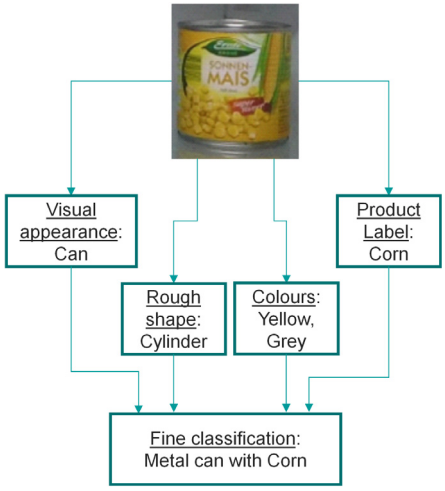


Fig. 3. Illustration of generic hierarchical concepts for fine object classification.

Table 1
Classes for the category 3D shape.

Name	category_id
Cup	1
Flat	2
Bottle	3
Cube	4
Orb	5
Tube	6
Irregular	7
Cylinder	8
undefined	9

shape of the object, independent of the current view. A slightly more specific category, denominated Visual category (Table 2), aims instead at classifying the object into various general food and beverage classes, but only taking into account its visual appearance and not trying to infer any other properties of the object's content. Finally, the higher level concept has been denominated Logical category (Table 3), and it aims at providing a general information about the object going beyond its outer visual appearance and inferring its content. In order to easily augment the annotations, and to provide more low-level information, three additional categories have been defined, also shown in Table 3, which are directly linked to the Logical category:

- Logical group: a general grouping of objects based on their food or beverage classes;
- Consistency: definition of the object's content consistency;
- Fine shape: a finer definition of the object's shape, especially useful for fruit and vegetables, *undefined* if not directly linked to a Logical category.

The image segmentation and object classification is performed through a conventional Mask R-CNN, He et al. (2017), with a Resnet50 backbone, He et al. (2016). All models have been implemented in Python, employing the *Pytorch* and *Torchvision* frameworks, Paszke et al. (2019). In order to take advantage of the hierarchically defined categories, however, the standard Mask R-CNN architecture has been slightly modified, as in Fig. 4. In fact, instead of using the common approach for multiple-labels classification through binary attributes, it has been decided to treat the classification objective as the parallel classification of all different categories. For this purpose, the final layer of the Mask R-CNN's region of interest (ROI) head, which in the conventional implementation is a linear layer whose outputs are the classes log-likelihood vector (also known as logits), has been modified as a list of linear layers, each one outputting the likelihood vector of

Table 2
Classes for the Visual category.

Name	category_id
Cup	1
Can	2
Transparent packet	3
Eggs	4
Jar	5
Bottle	6
Fruit or Vegetable	7
Dish	8
Cheese	9
Net	10
Box	11
Tetrapack	12
Tube	13
Tupperware	14
Opaque package	15
undefined	16

the categories in use. In this way, for each bounding box and mask, a list of likelihood vectors is obtained, all referring to the same object. This choice has the advantage of automatically avoiding incompatible combinations which can occur in standard binary multi-labels (eg., 3D shapes Cube and Bottle at the same time), while providing a way for recognizing ambiguous situations (eg., a Bottle, depending on the view, might be seen also as Cylinder, and this is a less serious error than classifying it as Cube).

The Mask R-CNN has been trained to predict only the manually-defined low-level tags (3D shape and Visual category) and the Logical category. The model has been initialized with the weights pre-trained on Imagenet, and the parameters for the first residual blocks group were kept frozen and not retrained. The optimizer was Adamax, Kingma and Ba (2017), with initial learning rate 10^{-4}, and the training data has been randomly sampled in batches of size 7 for 20 training epochs. The choices of the optimizer and of the training hyperparameters have been performed on the validation set. The classification loss function is modified to be the sum of conventional cross-entropy losses calculated separately on each of the different N categories:

$$L_{class} = \sum_{i=1}^{N} \sum_{k=1}^{K} \frac{1}{K} \left(-\log \frac{\exp\left(x^i_{k,y^i_k}\right)}{\sum_{c=1}^{C^i} \exp\left(x^i_{k,c}\right)} \right) \tag{1}$$

where K is the number of data samples, C^i the number of classes in category i, y^i_k the ground truth label for category i of object k and $x^i_{k,c}$ the calculated logit for class c of category i for object k. All the other components of the Mask R-CNN loss function are left as in the original implementation, and therefore the overall loss function to be minimized is given by:

$$L_{MRCNN} = L_{class} + L_{obj} + \lambda_{loss} \left(L_{boxr} + L_{rpn} \right) + \lambda_{mask} L_{mask} \tag{2}$$

where the losses components are the usual objectness loss (L_{obj}), box region loss (L_{boxr}), region proposal network loss (L_{rpn}) and mask loss (L_{mask}). The weight hyperparameters λ_{loss} and λ_{mask} have been tuned to 1.

3.3. Fine classification using a concept bottleneck

In order to make use of the hierarchical categories structure defined in the previous section, a model derived from the Concept Bottleneck approach of Koh et al. (2020) has been developed, Fig. 5. The model consists in a series of modules:

- an Object Detection Mask R-CNN, which subdivides the original images into object cutouts, and predicts the lower-level concepts;

Table 3
Classes for the Logical category, and connection to its additionally defined categories.

Logical category	Logical group	Consistency	Fine shape	category_id
Apple	Fruit	Hard	Round	1
Eggplant	Vegetable	Hard	Elongated Cone	2
Baking Ingredient	Carbs/Bakery	Soft	*undefined*	3
Banana	Fruit	Hard	Arked Cylinder	4
Beer	Liquids	Liquid	Bottle	5
Bread	Carbs/Bakery	Hard	*undefined*	6
Butter	Dairy	Hard	Elongated Cube	7
Eggs	Proteins	Hard	Oval	8
Ice Cream	Dairy	Soft	*undefined*	9
Premade food	Packaged food	Hard	*undefined*	10
Fish	Proteins	Hard	Irregular	11
Meat	Proteins	Hard	Irregular	12
Fruit Juice	Liquids	Liquid	*undefined*	13
Vegetable	Vegetable	Hard	*undefined*	14
Beverage	Liquids	Liquid	*undefined*	15
Spices/Herbs	Preserves	Hard	*undefined*	16
Cucumber	Vegetable	Hard	Cylinder	17
Pickles	Preserves	Hard	Cylinder	18
Homemade closed food	Homemade food	Hard	undefined	19
Homemade open food	Homemade food	Hard	undefined	20
Snack	Packaged food	Hard	*undefined*	21
Yogurt	Dairy	Soft	*undefined*	22
Cheese	Dairy	Hard	*undefined*	23
Coffee	Liquids	Liquid	Tall Cup	24
Carrot	Vegetable	Hard	Thin Cylinder	25
Potato	Vegetable	Hard	Elongated Round	26
Cabbage	Vegetable	Hard	Irregular Round	27
Coconut	Fruit	Hard	Round	28
Preserve	Preserves	Hard	Cylinder	29
Cake	Carbs/Bakery	Hard	*undefined*	30
Pumpkin	Vegetable	Hard	Elongated Round	31
Corn	Vegetable	Hard	Cylinder	32
Jam	Preserves	Soft	Cylinder	33
Melon	Fruit	Hard	Round	34
Milk	Dairy	Liquid	*undefined*	35
Dairy Product	Dairy	Soft	*undefined*	36
Fruit	Fruit	Hard	*undefined*	37
Oil	Liquids	Liquid	Bottle	38
Orange	Fruit	Hard	Round	39
Pepper	Vegetable	Hard	Bell Cylinder	40
Pasta	Carbs/Bakery	Hard	Irregular	41
Salad	Vegetable	Hard	Irregular	42
Sauce	Preserves	Soft	*undefined*	43
Soft drink	Liquids	Liquid	*undefined*	44
Asparagus	Vegetable	Hard	Thin Cylinder	45
Spirit	Liquids	Liquid	Tall Bottle	46
Sweets	Packaged food	Hard	*undefined*	47
Tea	Packaged food	Hard	*undefined*	48
Dough	Carbs/Bakery	Soft	*undefined*	49
Tomato	Vegetable	Hard	Collection Round	50
Grape	Fruit	Hard	Collection Round	51
Water	Liquids	Liquid	Transparent Bottle	52
Wine	Liquids	Liquid	Tall Bottle	53
White meat	Proteins	Hard	Irregular	54
Sausage	Proteins	Hard	Arked Cylinder	55
Onion	Vegetable	Hard	Round	56
undefined	*undefined*	*undefined*	*undefined*	57

- a Content Segmentation Mask R-CNN and Colour extractor. The content segmentation model divides each object into three regions: label, content and package. For each of these regions the colour extractor derives a colour histogram in the L*a*b space, taking as anchors the 16 colours: dark salmon, brown, dark orange, red, yellow, green, forest green, cyan, steel blue, blue, dark violet, magenta, white, gray60, gray40, black. Examples are shown in Fig. 6;
- a classifier, specifically a multi-layer perceptron (MLP), which takes all the lower-level concepts and colour histograms as input and predicts the Logical category for each object.

All these models have been implemented in Pytorch, and both Mask R-CNNs make use of the Resnet50 backbone. The Object Detection Mask R-CNN is trained in exactly the same way as in Section 3.2, but omitting the prediction of Logical category and instead additionally predicting the categories automatically derived from it, defined in Table 3 (Logical group, Consistency and Fine shape). On the other hand, the Content Segmentation Mask R-CNN has again the same structure of Section 3.2 and Fig. 4, with only one category as output, but the same Resnet50 backbone pre-trained on ImageNet. However, in this case, the network has been re-trained on a specifically defined and annotated dataset, derived from image cutouts of objects extracted from the main database. Finally, the classifier is a multi-layer perceptron with one hidden layer and a ReLu non-linearity. Its input is a 104-dimensional vector that contains the class logits ($16+9+17+4+10 = 56$ features from the logits of the five categories) and colour histograms ($3 * 16 = 48$ features from the colours on the three content regions) extracted for each detected object. Note that the Object Detection Mask R-CNN and Content Segmentation Mask R-CNN share a very similar backbone, in

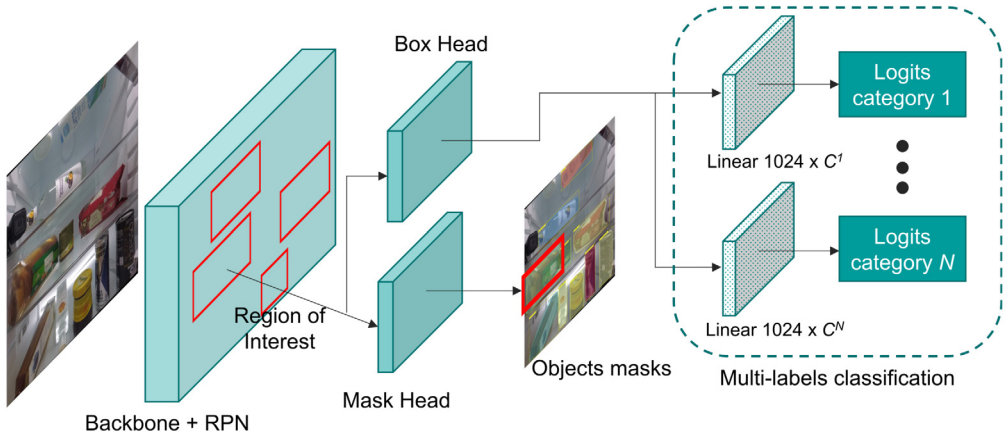


Fig. 4. Diagram illustrating the Mask R-CNN architecture, comprehensive of the proposed multi-labels classification block. The number of features from the final linear layer of Resnet50 is 1024, and these are combined through a set of linear layers to obtain the logits of each of the C^i classes in category i.

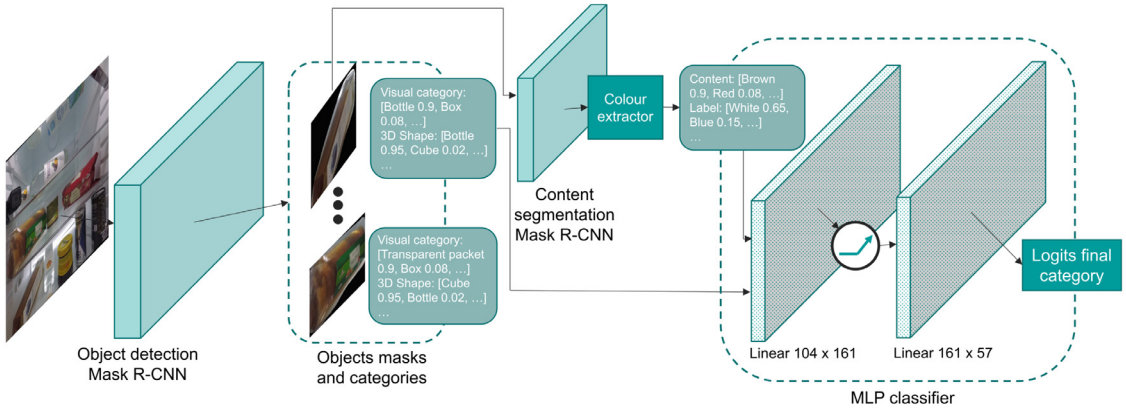


Fig. 5. Diagram illustrating the whole architecture of the Concept Bottleneck model used for fine-classification.

which the parameters for the first residual blocks are kept frozen to the pre-trained ones provided by PyTorch. On the other hand, all the other layers have been separately retrained on the two specific datasets, and the classification task is multi-label for the Object Detection Mask R-CNN, while single-label for the Content Segmentation Mask R-CNN.

Training has been performed on each of the modules separately. Specifically, in all cases Adamax was used, with initial learning rate 10^{-4} for the Mask R-CNNs and 10^{-3} for the classifier. The Mask R-CNNs have both been trained for 20 epochs, and the classification loss function is either a regular cross-entropy (for the MLP classifier and the Content Segmentation Mask R-CNN), or the sum of cross-entropy losses on each tag, as in Eq. (1) (Object Detection Mask R-CNN). In order to assess the importance of the error on the objects detection in the performance of the final classifier, its training procedure has been carried out in two different ways:

1. the input lower-level concepts come from the ground-truth labels;
2. the input lower-level concepts are calculated with the Object Detection Mask R-CNN.

Given the different nature of the two scenarios, the number of training epochs for the classifier in case 1 was 1500 and in case 2 was 600. Moreover, since each data point consists of a single object, unlike the original images which contain a multitude of them, it has been decided to balance the classes. Indeed, as evident from Fig. 7, the number of examples per class for the Logical category can vary by orders of magnitude. It has then been chosen to keep a maximum of 40 examples per class during training, and these are chosen randomly at each training epoch.

To evaluate the results on fine classification, in addition to the classical evaluation metrics precision (p), accuracy, recall(r) and specificity, we show the mean Average Precision mAP and F_1 score. mAP (Eq. (4)) is calculated as the mean for the 11-point interpolated average precision, as defined for the Pascal VOC2008 dataset (Everingham et al., 0000) challenge. The formula for the F_1 score is presented in Eq. (4).

$$mAP = \frac{1}{11} \sum_{r\in\{0,0.1,0.2,\ldots,1\}} max_{\tilde{r}\geq r} p(\tilde{r}) \tag{3}$$

$$F_1 = 2 \cdot \frac{p \cdot r}{p + r} \tag{4}$$

In addition, we propose a modification to the conventional mAP and F_1 score that aims to show the differences in explainability. The first step is the calculation of the normalized Euclidean distance $d_{i,j}$ in the dataset between each pair of Logical categories i and j, represented as the concatenation of the one-hot encodings of all lower-level categories ground-truths (Visual category, 3D shape, Logical group, Fine shape, Consistency). Since different objects pertaining to the same Logical category can have different lower-level tags, for each pair of Logical categories the median distance is then calculated:

$$d_{i,j} = \text{median}_{k\in K} \left(\frac{1}{F} \sqrt{\sum_{f=1}^{F} \left(o_{i,f}^k - o_{j,f}^k \right)^2} \right) \tag{5}$$

where $\text{median}_{k\in K}$ is the median across all K combinations of objects from classes i and j in the dataset, $o_{i,f}^k$ is the one-hot encoding of the ground-truth features for object k in class i and F is the dimension of the o vectors. The results are shown in Fig. 8, and it can be verified

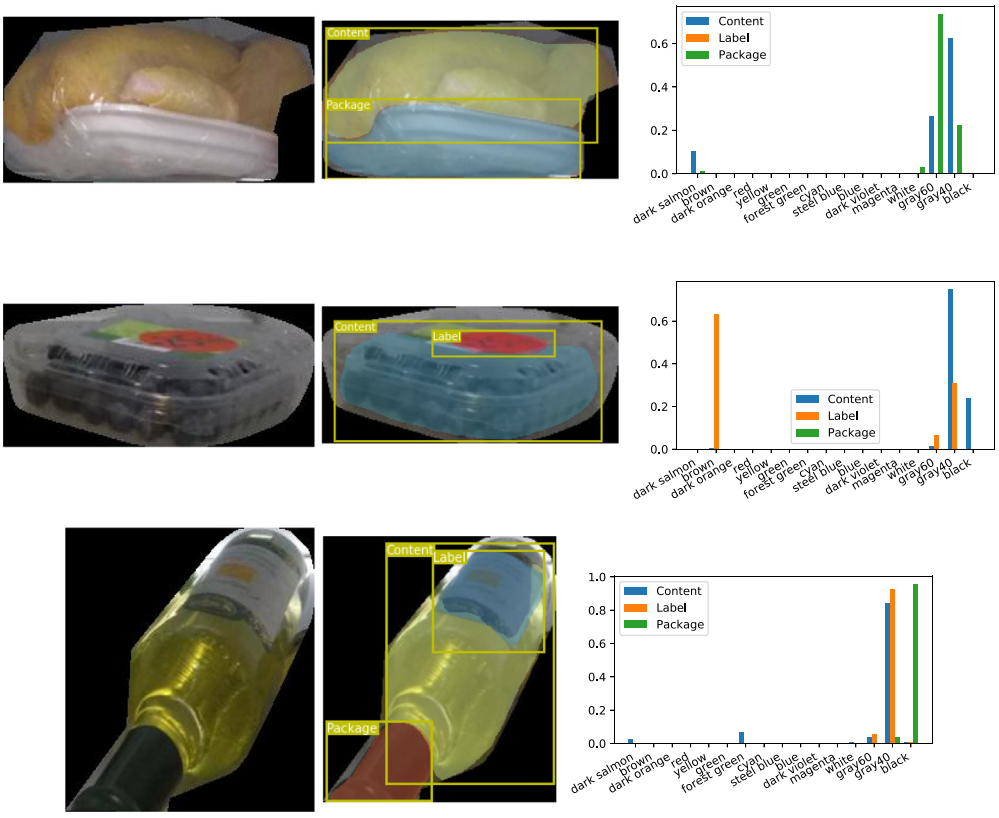


Fig. 6. Examples of the application of the Content Segmentation Mask R-CNN and Colour extractor on three objects shown on the left. The colour histograms for the three regions detected as content, label and package are shown on the right side.

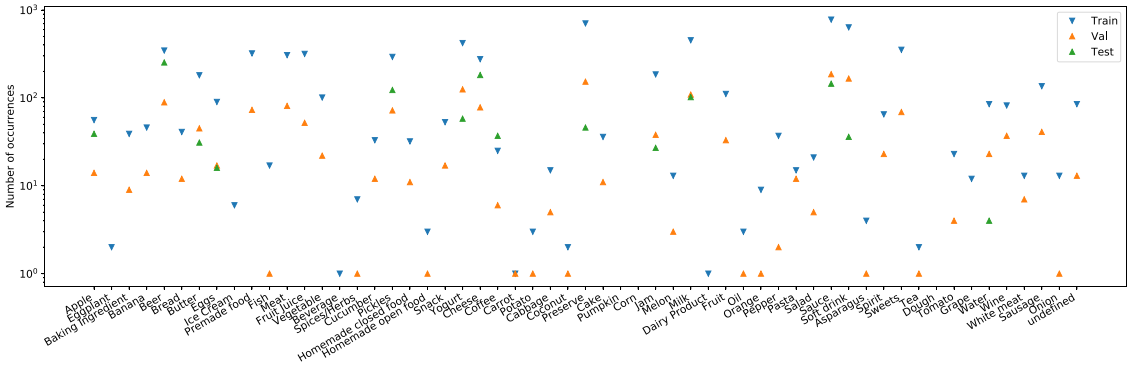


Fig. 7. Number of occurrences for each class in the Logical category.

that this distance is indeed representative of the logical distance between two categories. The final computation of an explainable metrics requires, then, a slight modification to the conventional calculation. In particular, a prediction is considered true positive not only if the predicted class is the same as the ground truth, but also if they are different, but the distance between the classes is below or equal to a certain threshold $d_{i,j} \leq d_{thr}$. As a function of d_{thr}, different values of mAP can be computed, essentially being equivalent to conventional mAP for $d_{thr} = 0$, and always reaching $mAP = 1$ for $d_{thr} = 1$. Similarly, the explainable F_1 score equals the conventional value for $d_{thr} = 0$, while $F_1 = 1$ for $d_{thr} = 1$.

3.4. Objects tracking

In order to demonstrate the high expressive power of the intermediate features derived in the Concept Bottleneck approach, their use for object tracking has been explored. In fact, in our target application, it is important to detect whether there have been any variations between two consecutive pictures of the fridge in terms of displaced, rotated, removed or inserted objects. To accomplish this, an object tracking model has been derived, to identify the relationships between each of the objects from figure i and the ones from figure $i + 1$. Incidentally, an accurate object tracker would also allow for the collection of multiple views of the same object, thereby enabling a better classification performance.

The developed Object Tracker, depicted in Fig. 9, takes as input the Concept Bottleneck features of the two object cutouts to compare, i.e., the low-level categories and the colours of the regions derived from the Content Segmentation Mask R-CNN. The features from the two images are then combined by calculating their absolute difference, are then concatenated with the Euclidean distance between the two locations in the original images and the absolute difference between the logarithms of their areas and are finally presented as input to the classifier. The latter is a binary classifier, which outputs the probability

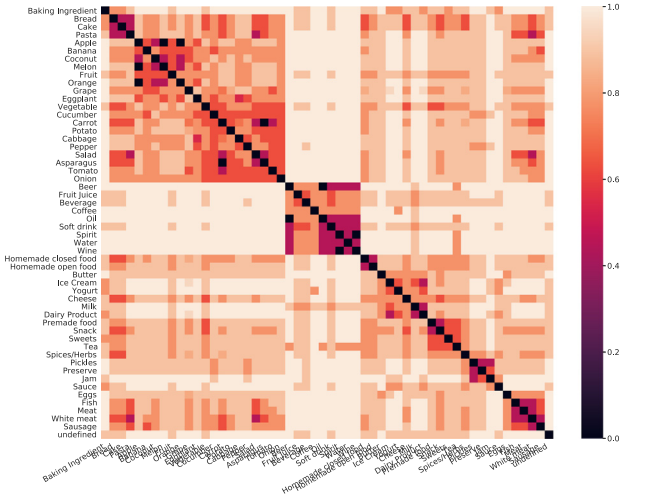


Fig. 8. Normalized Euclidean distances $d_{i,j}$ (Eq. (5)) between every object pertaining to each pair of Logical categories. The Logical categories have been ordered so as to match their Logical group, to show that logically similar objects indeed exhibit smaller distances.

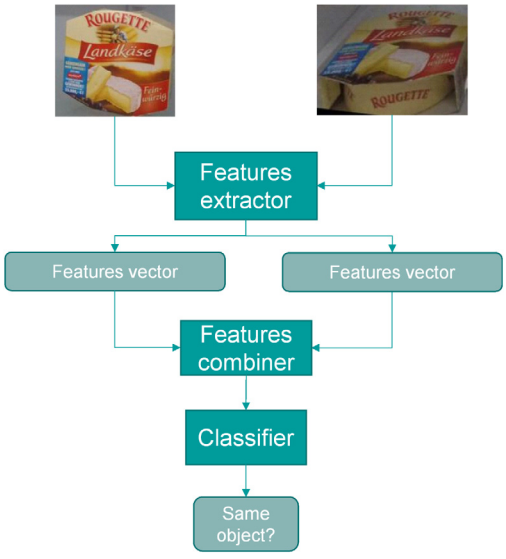
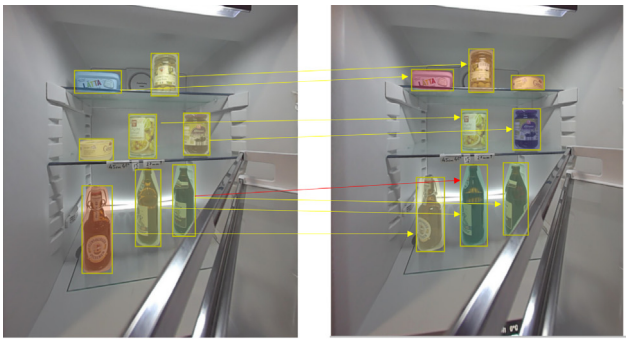


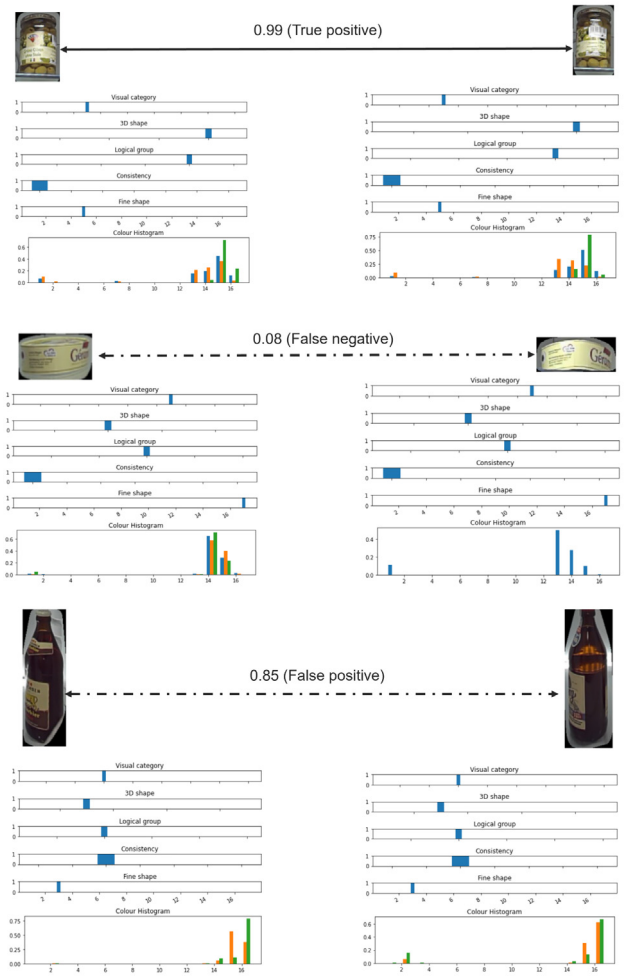
Fig. 9. Scheme of the proposed Objects Tracking models. The Features extractor module can be chosen in two different ways, producing either the Concept Bottleneck features or the output from the last linear layer of a Resnet50. In both cases, the features from the two images are then combined by calculating their absolute difference, and augmented with the distance and area ratio, as explained in the text.

of the two images being from the same object, and it is a multilayer perceptron with one hidden layer, a ReLu non-linearity and 106 input features. Only the classifier has been trained, keeping the Concept Bottleneck model frozen, using the Adamax optimizer for 500 epochs with an initial learning rate of 10^{-2}. The training and test datasets are the ones described as number 2. at the end of Section 3.1. The training data is also dynamically balanced at each step by keeping only a maximum of 5 negative examples for each positive one.

An example of the behaviour of the trained Object Tracker on two consecutive images from the test set is shown in Fig. 10. It can be observed that most objects are rotated and a few are displaced, but also that the point of view differs slightly between the images (Fig. 10(a)). In each image in Fig. 10(a) the objects detected via the Object detection Mask R-CNN are shown, together with the tracking output using the trained Object Tracker. Matched pairs between the detected objects in the two images are shown by arrows, where the yellow arrows show



(a) Object tracking output between two images.



(b) The input data and performance of the Object Tracker applied on 3 pairs of detected objects.

Fig. 10. Examples of object tracking between two consecutive images in the test set using the proposed model.

correct pairs (true positives) and the red arrows show false positive pairs. More details on the performance of the tracking model are shown on three pairs of objects detected in the two images (Fig. 10(b)). For each detected object in the two images, the Concept Bottleneck features (class logits and colour histograms) are shown. Moreover, the matching score and evaluation status obtained with the Object Tracker are given. A pair of detected objects is considered a match if the matching score is above the threshold $m_{thr} = 0.7$. The examples in Fig. 10(b) show true positive (detected objects are matched and belong to the same item), false negative (detected objects are not matched, but belong to the

Table 4
Results in terms of mAP for the Logical category prediction on the different sets, using a value of 0.5 for the mask intersection over union.

Model	mAP	Train.	Valid.	Test
Regular Mask R-CNN	No weights	0.75	0.76	0.82
	By classes freq.	0.97	0.97	0.90
Conc. Bottleneck on annotations	No weights	0.89	0.89	0.99
	By classes freq.	0.96	0.96	0.98
Conc. Bottleneck complete	No weights	0.74	0.69	0.77
	By classes freq.	0.91	0.90	0.90

same item) and false positive pair (detected objects are matched, but do not belong to the same item). The false negative pair is found in a scenario where the object was rotated and displaced on a different shelf, with a change in illumination. On the other hand, the false positive pair is found between two quite similar bottles of bier. The tracking error examples show the difficulty of tracking in the given test case. Specifically, the data have similar objects originating from the same fine-grained classes, and there may be a large change in the position and orientation of the objects between two consecutive images. For comparison with more traditional methods, another architecture of the Object Tracker has been developed. The only difference in Fig. 9 with respect to the previous case is the choice of the Features extractor, being in this case the last linear layer of a Resnet50 model, which provides an array of 1024 features per image. The features from the two images are then combined in exactly the same way as when using the Concept Bottleneck features, and again augmented with the distance and area ratio. The Resnet50 model is the same one which constitutes the backbone of the Object Detection Mask R-CNN, and it has been verified that fine-tuning it does not result in a measurable performance increase. Once again the Adamax optimizer has been used, running for 1400 epochs with an initial learning rate of 10^{-3}.

4. Results

4.1. Fine classification

In order to understand the influence of the algorithm on the accuracy and explainability of the predictions, three different models have been used:

- a regular Mask R-CNN trained on 3D shape, Visual category and Logical category, as described at the end of Section 3.2;
- a Concept Bottleneck model, with two different configurations, as described in Section 3.3:
  - the complete model, as represented in Fig. 5;
  - a model avoiding the use of the Object Detection Mask R-CNN, but rather using directly the ground-truth annotations for the hierarchical categories.

The results obtained with these three models on the prediction of the defined Logical category are summarized in Table 4, in terms of mAP on all classes for training, validation and test sets. Due to the large imbalance in the classes occurrences, in addition to the conventional mAP, also a weighted mAP is calculated, weighting the average between the classes with the relative frequencies of occurrence of the objects from that class. Since the validation and training sets are randomly sampled from the same images sequences, it is unsurprising that the mAP is very similar in most cases between the two sets. Noting the relatively large difference between regular and weighted mAP, it is also evident that there is a substantial impact from higher errors on rarely seen classes, which is especially prominent in the case of the Regular Mask R-CNN. The Concept Bottleneck model, on the other hand, suffers less from this imbalance, due to the balancing scheme implemented in the training procedure. On the other hand, the comparison between the two Concept Bottleneck models shows that, as expected, using the ground-truth annotations the accuracy is extremely high, almost perfect on the test set. This observation justifies the choice of features for the model, and shows that the greatest part of the model's error is due to the incorrect classification of the hierarchical categories from the Object Detection Mask R-CNN.

Moreover, the performance on the test set is also excellent, and the difference between mAPs on the validation and test sets is negligible for the Concept Bottleneck model, unlike for the Regular Mask R-CNN. This happens even though, unlike the validation set, the test set contains objects that were not part of the training set, yet are still similar to those used during training. The lower difference between mAP on validation and test sets shows that the Concept Bottleneck model is better at generalizing to unseen objects, that are though still extracted from the same distribution of the training set.

For a more detailed investigation of the results on the test set, Table 5 shows several evaluation metrics and Fig. 11 shows the confusion matrices, restricted only on the classes present in the test set. The results in Table 5 show that the accuracy and specificity are high for all models, while the greatest difference between the models is seen in precision and recall, which affect the F_1 and mAP (Table 4). The performance of the Concept Bottleneck on annotations is especially good, showing that the complete Concept Bottleneck model would also have excellent performance if the object detection and classification to the low level categories was improved. On the other hand, the performance of the complete Concept Bottleneck model is on par with the Regular Mask R-CNN. However, we will show that the Concept Bottleneck model does still provide advantages in terms of explainability of the predictions and in handling classes confusion.

Investigating in particular the confusion between classes, it can be noted in Fig. 11(a) that the Regular Mask R-CNN does not always lead to confusion between similar classes. For instance, there is large confusion between classes Eggs and Butter or Milk and between Jam and Premade food or Sauce, which are not logically or visually similar. On the other hand, Fig. 11(b) shows the result when using the Concept bottleneck model, with the objects attributes extracted from the annotations. As already noted, the accuracy is almost perfect on the test set. In fact, only two classes lead to some confusion:

- Apple, which can be confused with Melon, being both round and of similar colour;
- Preserve, which can be confused with Pickles, being both in a glass jar.

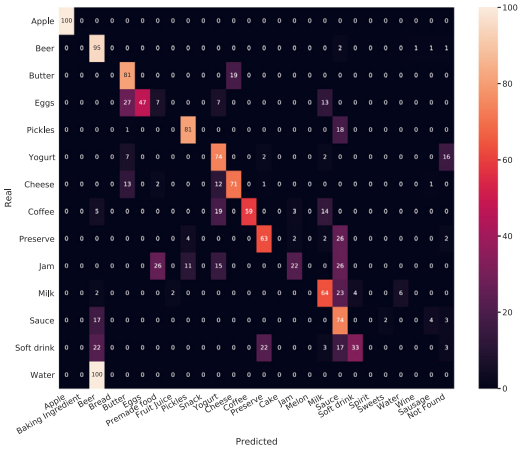
These results prove once again that, on the one hand, the defined lower-level concepts are expressive enough to allow for the prediction of the Logical category in most of the cases. On the other hand, it is also shown that, in a problem of very fine classification, it is unavoidable to have some confusion between classes, and that the domain can be easily partitioned in different regions depending on the particular choice of classes, where an overlap between regions cannot be prevented.

Finally, Fig. 11(c) shows the confusion matrix on the test set using the complete Concept Bottleneck model, including the Object Detection Mask R-CNN. Comparing these results to Fig. 11(b), it is clear the drop in accuracy, especially on the test set, due to not relying on ground-truth annotations. The comparison with Fig. 11(a), instead, shows that the accuracy on the test set is not substantially different in this case. The main differences lie in the confusion between similar classes, for instance:

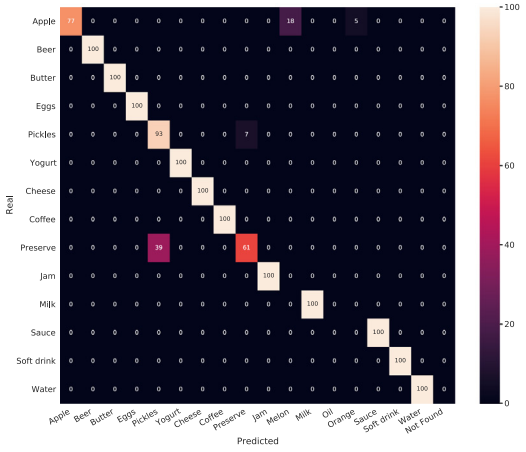
- Eggs, can be now confused with Baking ingredient or not found, while the regular Mask R-CNN confused them with Butter and Milk;
- Jam, can be now confused with Yogurt and Pickles, while previously it could be also confused with Premade food and Sauce;
- Soft Drink, can be now confused with Coffee and Sauce, while previously it was confused with Beer, Preserve and Sauce.

Table 5
Evaluation metrics for the Logical category prediction on the test set, using the different models.

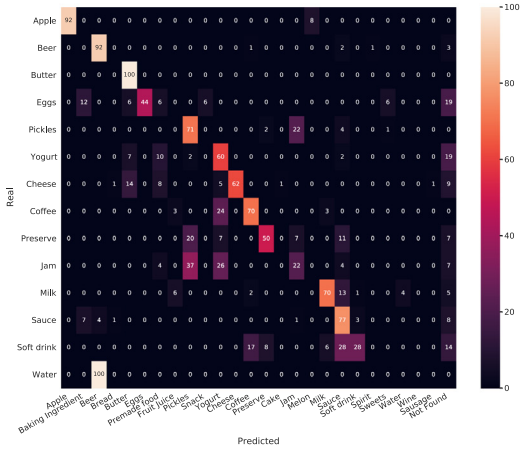
Model	Weight	Precision	Accuracy	Recall	Specificity	F_1
Regular Mask R-CNN	No weights	0.75	0.97	0.62	0.98	0.70
	By classes freq.	0.82	0.95	0.76	0.97	0.77
Conc. Bottleneck on annotations	No weights	0.97	0.995	0.95	0.998	0.96
	By classes freq.	0.97	0.996	0.97	0.998	0.97
Conc. Bottleneck complete	No weights	0.69	0.97	0.63	0.99	0.69
	By classes freq.	0.84	0.96	0.76	0.98	0.78



(a) Regular Mask R-CNN



(b) Conc. Bottleneck on annotations



(c) Conc. Bottleneck

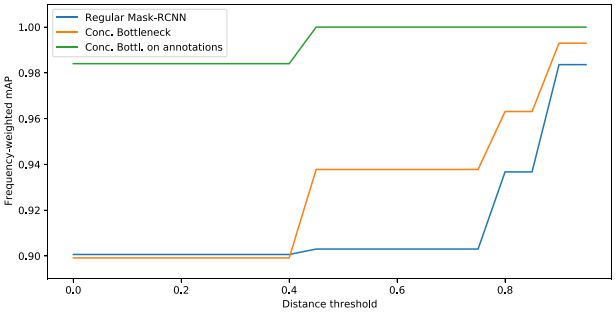
Fig. 11. Confusion matrices for the prediction of Logical category on the test set, using the different models. The matrix on the test set is not symmetrical for better readability, avoiding to show the empty lines of categories that are not in the annotations but have been predicted (e.g., Fruit Juice).

Overall, the observation is that, while the precision for each of the classes does not change significantly between Figs. 11(a) and 11(c), on the one hand, the explainability of the wrong predictions is now increased, and the confusion is clearly between similar classes. On the other hand, the Concept bottleneck has increased the number of Not found examples, which can be seen as beneficial in the cases in which the Regular Mask R-CNN would otherwise be wrongly predicted, but this is reflected in the slight drop in mAP and other metrics, observed accordingly in Tables 4 and 5.

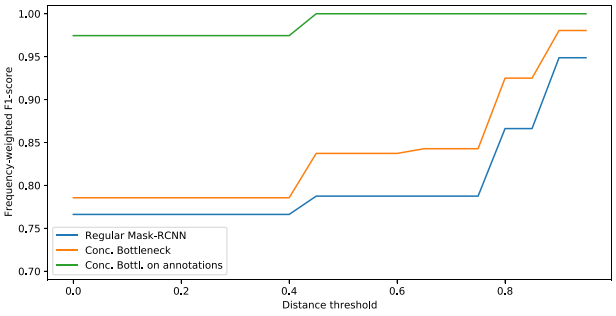
In order to investigate more quantitatively the increase in explainability provided by the Concept Bottleneck model, Fig. 12 shows the calculation of Explainable mAP and F_1 score (as defined at the end of Section 3.3) for the three different models on the test set, as a function of the categories distance threshold d_{thr}. It can be observed that, above $d_{thr} = 0.5$, the Concept Bottleneck model always has a higher value of Explainable mAP and F_1 score than the regular Mask R-CNN. This proves that the Concept Bottleneck more easily confuses different Logical categories if their distance from each other (defined by Eq. (5)) is low, i.e., if the two classes pertain to similar objects. On the other hand, the regular Mask R-CNN can more easily confuse classes that do not have high similarities to each other. Overall, Fig. 8 shows that $d_{thr} = 0.5$ is a value that can reliably differentiate between similar and distant classes.

4.2. Object tracking

The evaluation of the object tracking approach is performed on the four image sequences in the test set. For each image sequence, all matching pairs of detected objects are obtained in every two consecutive images, taking into account all combinations between object k from image i and all other objects from image $i + 1$. The matching pairs have a matching score above the threshold $m_{thr} = 0.7$. Fig. 13



(a) Explainable mAP



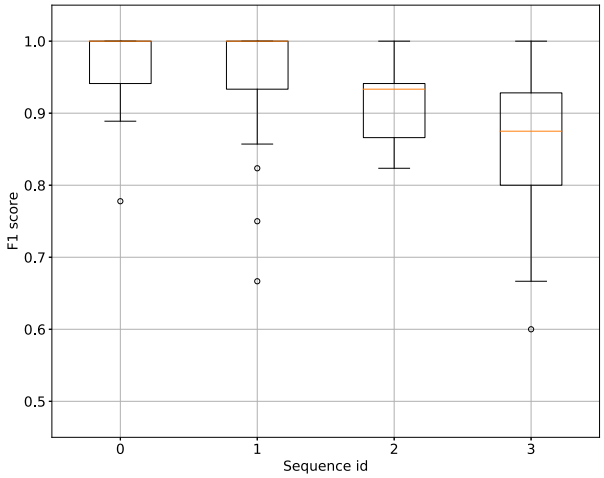
(b) Explainable F_1 score

Fig. 12. Explainable metrics on the test set as a function of the categories distance threshold d_{thr} for the three different models. The average between classes has been weighted by the classes occurrences frequencies.

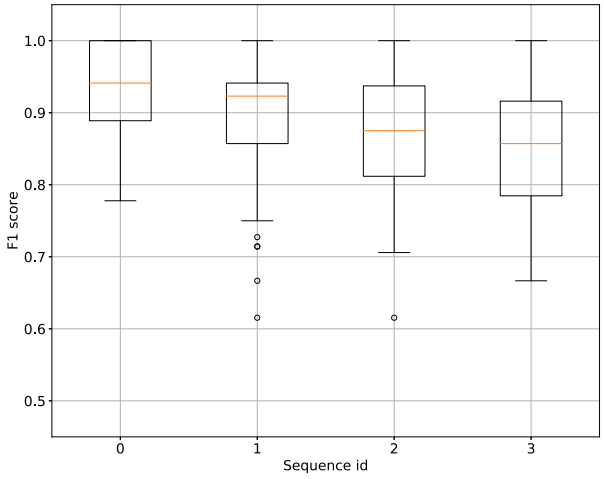
shows the results of the object tracking algorithm which makes use of the hierarchical features defined in the Concept bottleneck model. In the top plot, the features for the classifier are derived from the ground-truth annotations, while in the bottom plot these features are calculated by the Concept bottleneck model. It is evident that, also in this case, using the ground-truth annotations gives higher accuracy, however the difference is not as large as in the case of Logical category classification. Also in the worst scenario, the accuracy is usually high with a median F_1 score of around 0.85 in the sequence. The error is highly dependent on the sequence, and it is evident the effect of lower variability in sequences 0 and 1, resulting in a lower classification error.

On the other hand, Fig. 14 shows similar results to Fig. 13, with the only difference being the use of the Resnet50 features instead of the ones coming from the Concept bottleneck. Two effects are immediately evident, one being the larger variability of the error between different images in the sequence, signalled by the greater height of the boxes. The second effect is the decrease in median F_1 score in the sequence with respect to Fig. 13, demonstrating that the Concept bottleneck features can be trained to be more expressive towards identifying similarities between objects. One last effect is the slight increase in the variability between Fig. 14 top and bottom, the only difference being the way the image cutout is derived, in the former case coming from the annotations and in the latter being calculated by the Mask R-CNN. The higher accuracy and lower variability in the latter case is only explainable by the fact that, since the segmentation is performed with the same Resnet50 backbone that is then used to extract the features, the way the objects masks are derived in this case are more representative of the appearance of the objects.

For a more detailed comparison between the models, Table 6 shows a summary of the evaluation metrics in all cases. Besides the F_1 score, two other metrics defined in Bernardin et al. (2006) for multiple object tracking are calculated: multiple object tracking accuracy $motA$ and multiple object tracking precision $motP$. The first metric accounts for all errors made by the object tracker: the number of false negatives



(a) Concept Bottleneck features, Ground truth annotations



(b) Concept Bottleneck features, From Object Detection Mask R-CNN

Fig. 13. Boxplots of the F_1 scores for objects tracking between consecutive images for the 4 sequences on the test set. The features from the Concept bottleneck have been used for the image similarity classification, either using directly the ground truth annotations (top) or the ones calculated by the model (bottom).

objects or misses m, the number of false positives objects that are wrongly detected fp and the number of mismatches mme that include the wrongly matched pairs and truth pairs that are not matched. The metric $motA$ is calculated as:

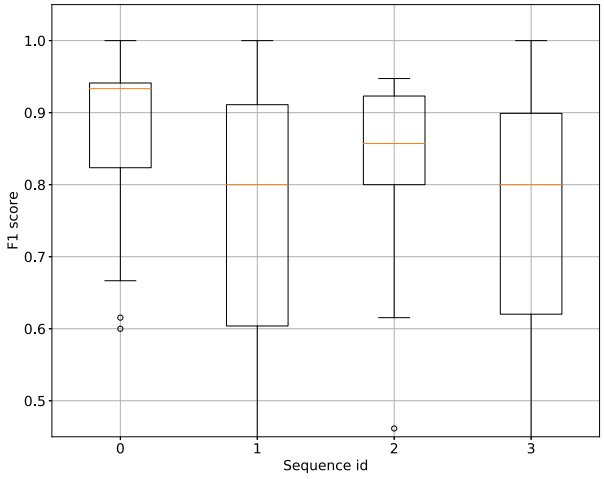
$$motA = 1 - \frac{\sum_i m_i + fp_i + mme_i}{\sum_i gt_i} \tag{6}$$

where i is the image index and gt_i is the true number of objects in image i. The comparison between F_1 and $motA$ shows that, as already noted, the models that use the Concept Bottleneck features have better results, compared to those that use the Resnet50 features.

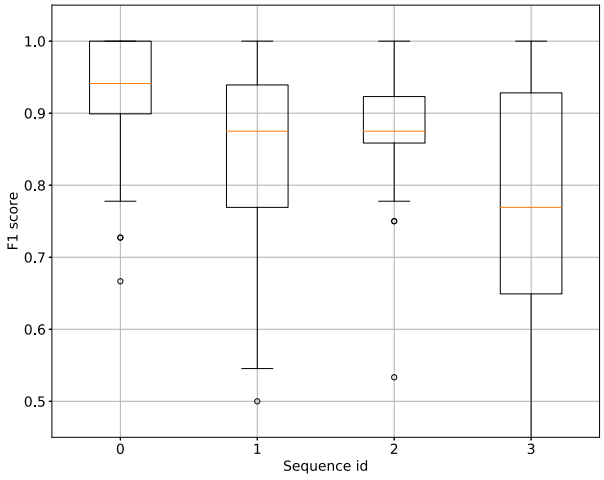
On the other hand, the last metric considered here, $motP$, shows the ability of the tracker to detect the objects and precisely determine their position. It is calculated as the ratio between the total position errors of the detected and ground truth object regions and the number of correctly detected objects. This metric is therefore independent of the choice of the features to use for the object tracking, and therefore cannot be used for comparing the two approaches. The total position error is obtained as the Euclidean distance between the centres of the detected and ground truth (annotated) region, where the height

Table 6
Evaluation metrics for object tracking on the four sequences in the test set using different features and annotations on object cutouts.

Annot.	Features	Sequence 1			Sequence 2			Sequence 3			Sequence 4		
		F_1	motA	motP	F_1	motA	motP	F_1	motA	motP	F_1	motA	motP
Ground truth	Resnet50	0.78	0.64	0	0.64	0.46	0	0.76	0.59	0	0.73	0.5	0
	Bottleneck	0.97	0.94		0.96	0.92		0.90	0.80		0.86	0.73	
Mask R-CNN	Resnet50	0.94	0.89	3.1e−3	0.85	0.72	3.7e−3	0.87	0.74	1.8e−3	0.78	0.52	2.1e−3
	Bottleneck	0.94	0.87		0.90	0.81		0.87	0.75		0.83	0.65	



(a) Resnet50 features, Ground truth annotations



(b) Resnet50 features, From Object Detection Mask R-CNN

Fig. 14. Boxplots of the F_1 scores for objects tracking between consecutive images for the 4 sequences on the test set. The features from the last linear layer of Resnet50 have been used for the image similarity classification, either using the image cutouts from the ground truth annotations (top) or the ones segmented from the Objects detection Mask R-CNN, part of the Concept bottleneck model (bottom).

and width of the images are previously scaled to 1. The results from Table 6 show that the position error is very low for the objects detected by the Mask R-CNN. Note that the models that use the ground truth annotations do not have any position error, by definition.

5. Conclusions

In this work, we have investigated the use of Concept Bottleneck models for fine classification and objects tracking, discussing the advantages of the approach with respect to classical black-box models. In particular, for fine classification of objects, the Concept Bottleneck approach has allowed better explainability of the predictions and handling of cases in which confusion can be unavoidable due to ambiguities between similar classes, with a minimal drop in mAP with respect to a conventional Mask R-CNN. A new method of calculating an Explainable mAP has also been proposed, that could quantitatively demonstrate the increase in explainability of the Concept Bottleneck model. Moreover, it has also shown its flexibility, being easily adapted for tracking objects in a sequence of images, obtaining a higher accuracy than more traditional methods.

Such a modular approach allows for easily designing improvements to the model. In future work, it will be investigated how to integrate in the concepts the information derived by reading the physical objects labels from the segmented regions via OCR methods. It will be also investigated the use of the object tracking model for obtaining different views of the same object, whose features can be combined to increase the classification accuracy, mitigating the problems of occlusion and bad visibility.

As with the original Concept Bottleneck, which was show-cased on both medical images and on birds classification, the methods proposed in this paper are fully general. In any application in which hierarchical partitioning of categories can be defined, our proposed Concept Bottleneck approach can be easily adapted for images segmentation and object tracking. The only component more tailored to our application is the Content Segmentation Mask R-CNN, which would need to be eliminated or modified in other applications. The code is released as open source and available at https://opensource.silicon-austria.com/pittinof/hierarchical-concept-bottleneck.

CRediT authorship contribution statement

Federico Pittino: Conceptualization, Methodology/Study design, Software, Validation, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision. **Vesna Dimitrievska:** Methodology/Study design, Software, Validation, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Rudolf Heer:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Federico Pittino reports financial support was provided by Silicon Austria Labs GmbH Austrian Smart Systems Integration Research Center.

Data availability

The authors do not have permission to share data.

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